

PET final-design selection study

Study deck (generated from committed results)

PET final-design study

PET is diagnostic method development. Nothing here is a publication adoption; no historical threshold, verdict or the predecessor campaign's disposition changes; simulation only.

Question, preference and decision rule

- ▶ Select one complete PET design (detector/truth representations, model and initialization, recipe, miss rule, iteration count, uncertainty procedure) among a declared, tested family — or conclude “no eligible design” / “unresolved” with the discriminating evidence.
- ▶ The smaller network is preferred *only* if non-inferior (simultaneous one-sided 95 %, margin -0.02 on the primary recovery, plus frozen regional/topology margins) *and* at least $2\times$ cheaper in measured total cost; otherwise the stronger larger/pretrained design.
- ▶ Numeric decision table frozen before any successor run (PROTOCOL-20260925.md); final inference conditional on frozen event banks; final-bank runs blinded until an explicit UNBLIND amendment.

Fresh-event capacity and independence

bank	rows	use
DEV (exposed)	45089191	development; every prior
FB (never drawn)	2646891	final pseudodata; sealed except listed runs
RB (pool R, never opened)	1414846	reserve; sealed

Only ≈ 3.4 historical-size replicates of never-drawn events exist and no further same-model simulation: final replicates are independent random subsets *given the banks*; the bank's own offset is reported beside every contrast.

Where the multiplicity failure sits: the detector step

Injected E_{avail} L1: tilt 0.273, protons $\times$ 1.3 0.026, neutrons $\times$ 1.3 0.011 — the multiplicity cases barely move the scored marginal.

	det. ΣE	$k=1$	$k=3$	$k=10$	pull p -class	$k=1$	$k=3$	$k=10$
CTL		-0.16	-0.37	-0.39		+0.01	+0.02	+0.05
C		-0.16	-0.45	-0.44		+0.01	+0.03	+0.08
B		+0.81	+0.82	+0.83		+0.11	+0.22	+0.49

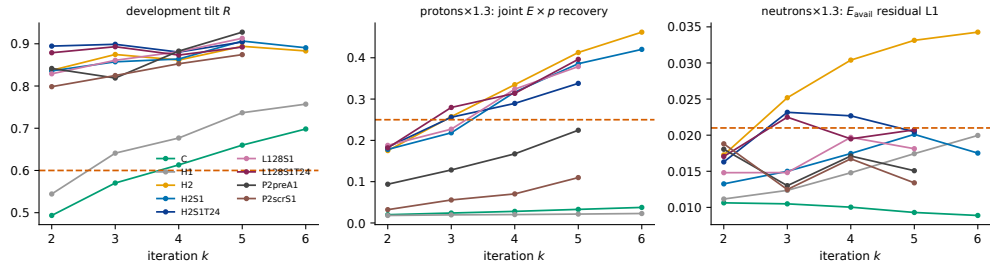
Protons $\times$ 1.3 (2 replicates): the baseline detector step (CTL, C) mis-closes the cluster-energy distribution and transmits almost no proton-content change to the pull; with reco summaries (B) both are transmitted. Development evidence.

Truth-step learnability (bounded interventions)

truth arm (step 2 on the exact weights)	p-class sel.	p-class missed	n-class sel.
raw PDG	0.83	0.83	0.59
one-hot PDG	0.87	0.88	0.88
one-hot + counts	0.89	0.92	0.88
one-hot, 24 epochs	0.97	0.96	0.98

Given C's real iteration-1 pull, the one-hot truth step recovers 0.012 of the joint $E_{\text{avail}} \times \text{proton change}$; given B's, 0.114. The truth step learns and extrapolates what it receives; the loss is upstream.

Development screen (DEV bank)



Reco summaries lift the tilt (H2S1 $k=5$: 0.906; C $k=3$: 0.570); the forced-1e-5 schedule fails the neutron screen (H2 $k=3$: 0.025) while the constant rate passes (H2S1 $k=5$: 0.020, topology 0.386). Development evidence, 2 draws per case.

Reproducibility repair (single-unfolding seed spread)

design	truth step	k	seed sd of R_{E0}	N2 (≤ 0.05)
H2S1	8 epochs	5	0.095	fail
L128S1	8 epochs	5	0.091	fail
L64S1	8 epochs	4	0.057	fail
CS1	8 epochs	6	0.054	fail
H2S1E16	16 epochs (both steps)	4	0.074	fail
L128S1E16	16 epochs (both steps)	4	0.071	fail
H2S1T24	24 epochs	5	0.037	pass
L128S1T24	24 epochs	4	0.043	pass

DEV draws $0-1 \times 4$ estimator seeds at fixed events. The spread sits in the truth step (the detector step varies little with the seed); a 24-epoch truth step repairs it (Amendment 3b).

Finalist rule applied mechanically

design	K^*	dev. tilt at K^*	S-N1	seed sd ($S-N2 \leq 0.05$)
H2S1T24	5	0.904	PASS	0.037
H2S1	5	0.906	PASS	0.095
H2S1E16	4	0.881	PASS	0.074
CS1	6	0.680	PASS	0.054
L128S1T24	4	0.873	PASS	0.043
L128S1	5	0.913	PASS	0.091
L128S1E16	4	0.875	PASS	0.071
L64S1	4	0.882	PASS	0.057
P2preS1	4	0.905	fail	—
P2preA1	—	—	—	—
P2scrS1	—	—	—	—

Rule committed before the recoveries it ranks were inspected, with addenda (completeness, S-N1, S-N2); finalists: compact H2S1T24 $K=5$, large L128S1T24 $K=4$ (Amendments 3c, 3d).

Algorithmic alternative: AUSSIE (scalar, matched)

matched pair	AUSSIE moves-away	OmniFold	AUSSIE worst R	OmniFold
AUSSIE lambda=1000 vs OmniFold carry-misses	11	6	-1.077	-0.217
AUSSIE lambda=1000 vs OmniFold carry-misses	11	6	-1.077	-0.246
AUSSIE lambda=1000 vs OmniFold carry-misses	11	6	-1.077	-0.246
AUSSIE lambda=1000 vs OmniFold carry-misses	8	6	-0.985	-0.230

Bounded matched stress test (same inputs, miss handling, data and tuning opportunity): AUSSIE loses on robustness and stability; it does not advance to a PET backbone.

Frozen finalists and final-stage design

- ▶ Compact **H2S1T24** ($K=5$) and large **L128S1T24** ($K=4$): detector reco summaries, categorical truth PDG, constant rate, 24-epoch truth step; step-1 PET 47 k vs 0.97 M parameters. Anchors CTL and C at $k=3$.
- ▶ Cost per unfolding (charged A100-h, declared packing): 1.77 (compact) vs 1.49 (large): the smaller network is not cheaper, so the cost-based preference cannot apply.
- ▶ Independent sizing: $n_F = 60$ (capped; E_0 non-inferiority between the finalists would need 250 draws — a quantified limit); D4c/D3 draws 40.
- ▶ Final bank: FINAL, 21-case library and coverage run blinded until the UNBLIND amendment.

Look 1: finalists on the final bank

look 1 (mean, simultaneous LB)	H2S1T24 $K=5$	L128S1T24 $K=4$
E_0 development tilt (U1)	0.895 (0.876)	0.863 (0.842)
moderate / good region (U2b)	0.874 / 0.917	0.846 / 0.910
E_3 opposite tilt (U3)	0.835 (0.815)	0.829 (0.809)
E_4 proton topology (U4)	0.391 (0.368)	0.348 (0.326)
$E_5 E_{\text{avail}} \times q_3$ (U5)	0.446 (0.429)	0.450 (0.435)
N2 seed sd (≤ 0.05)	0.037	0.043
gain over CTL in R_{E_0}	+0.589 (0.568)	+0.557 (0.535)
cost per unfolding (A100-h)	1.77	1.52
status (rules)	INCOMPLETE (C1-C5)	INELIGIBLE (B2)

Ranking: CONTINUE (candidates pending: H2S1T24K5). No look 2 (no sequential rule returned CONTINUE). Coverage of H2S1T24 runs blinded.

Look 1: what the verdict does and does not show

- ▶ **B2** (D4d n down, mean residual — injected E_{avail} L1, limit 0.010, a point rule): H2S1T24 0.0097 [0.0025, 0.0169] PASS; L128S1T24 0.0122 [0.0064, 0.0181] FAIL. **Point-decided, statistically unresolved**: both intervals cross the limit; the difference is not resolved by this comparison. Not evidence that the compact design is more robust, nor that the two are equivalent.
- ▶ **B1** PASSes under its frozen independence-based bound. Under within-draw dependence the evidence is insufficient to establish a per-unit failure probability ≤ 0.10 : on the common panel (FB0–7) H2S1T24 has 2/8 draws with a failure (CP upper 0.651), L128S1T24 1/8; with none the bound would be 0.369. This does not show the probability exceeds 0.10.
- ▶ Like-for-like and FB-population recoveries agree on the designated endpoints (report §6, report only).

Uncertainty calibration: coverage of H2S1T24

H2S1T24 $K=5$, $B = 6$, 120 replicates	measured	verdict
C1 pooled 95 % (LB ≥ 0.90 , point ≤ 0.99)	0.990 (LB 0.980)	unresolved
C1 pooled 68 % (LB ≥ 0.60 , point ≤ 0.80)	0.896 (LB 0.858)	decisive
C1 (both levels)		FAIL
C2 every bin 95 % ≥ 0.85	all bins pass	PASS
C3 moderate / good 95 % LB ≥ 0.85	0.949 / 0.993	PASS
C4 half-width / limit, bins 1–6 (≤ 1)	1.32–1.72	FAIL
C5 D4c up	skipped after decisive C1–C4 FAIL	—

In aggregate the six-member interval is **too wide**; the same against the FB population target. Not uniformly conservative: the low-acceptance region **under-covers** (68 %: 0.26, 95 %: 0.79), a region no rule gates. Mechanism not established.

Terminal outcome (frozen rules)

NO_ELIGIBLE_DESIGN

finalist	status (rules)
H2S1T24K5	INELIGIBLE (C1, C4)
L128S1T24K4	INELIGIBLE (B2)

- ▶ No design selected and no default named (the default must itself be eligible).
- ▶ Supported: H2S1T24's point estimator passes every accuracy, robustness and stability rule; its interval procedure fails calibration (too wide in aggregate; under-covers at low acceptance). Not supported: a ranking of the finalists.
- ▶ Remaining route (owner decision): \$11 repair — interval calibrated on DEV, re-validated on the sealed RB bank.

What this study cannot authorize

- ▶ No publication adoption; no real-data unfolding; no $C_{\text{stat}}/C_{\text{ML}}$; no Gate-6 work; OI-126 not reopened; the scalar-5D covariance unchanged; no note/primer/paper change.
- ▶ Historical thresholds and verdicts (`NEITHER_ELIGIBLE` / `NO_SELECTION`) and the predecessor's disposition are unchanged.
- ▶ Scope of any selection: signal-only simulation, the inventory's simulated detector response, the study's event scale, conditional on the frozen banks; generator dependence beyond histogram reweightings is unprobed.